MONEYGO

A MICRO PROJECT REPORT

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ABSTRACT

MoneyGo is a lightweight banking transaction program designed to simulate essential functionalities of a banking system using the Java programming language. The project is structured into two primary components: a **Customer Terminal** and a **Banking Interface**.

The **Customer Terminal** allows users to view their account details, including balance, transaction history, and account information, all retrieved from a dynamically updated .txt file. This file acts as a simple data store and is modified in real-time to reflect changes made during program execution.

The **Banking Interface** facilitates money transfers between accounts. When a transaction is executed, the respective balances in the .txt file are updated accordingly. The system ensures that data remains consistent and reflects real-time banking behavior.

To introduce a layer of security, **basic hashing** is incorporated. Sensitive data such as user passwords and bank account informations are hashed using a one-way hash function (e.g., SHA-256) before being stored or compared. This ensures that raw credentials & data are never saved in plain text.

The project showcases core programming concepts in Java, such as:

- Object-Oriented Programming (OOP)
- File I/O operations
- String parsing and data formatting
- Basic hashing
- Input validation and error handling
- Modular design and a console-based user interface

MoneyGo is designed as a beginner-friendly educational project, offering practical exposure to how simple financial systems function behind the scenes. It lays a solid foundation for more advanced features in the future, such as database integration, GUI development, and secure authentication mechanisms.

Customer Terminal – Features Table

Feature	Description
View Account Details	Reads from a .txt file and displays user details like name, account number, balance.
Transaction History	Displays recent transactions related to the logged-in user (read from file).
Password Protection	Login protected via hashed password stored in the file.
Hash Verification	User-entered password is hashed and compared with stored hash.
Input Validation	Basic checks for input format, account number, etc.
Real-Time Updates	Reflects updated balance and transactions as changes occur.

Banking Interface – Features Table

Feature	Description
Transfer Money	Initiates a transaction between two user accounts.
Update File Data	Modifies .txt file to update sender and receiver balances.
Record Transaction	Appends transaction details (amount, date, accounts) to transaction log file.
Hashing	Optionally hash each transaction log line to prevent tampering (basic integrity check).
Account Creation	Allows admin or bank staff to create new accounts with hashed passwords.